

SI 010 CNV

Installation and operating instructions

Template subtitle



Declaration of conformity

GB Declaration of Conformity

We, Grundfos, declare under our sole responsibility that the product SI 010 CNV, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

- Low Voltage Directive (2006/95/EC).
Standard used: EN 61010-1: 2001.
- EMC Directive (2004/108/EC).
Standard used: EN 61326-1: 2006.

CZ Prohlášení o shodě

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobek SI 010 CNV, na nějž se toto prohlášení vztahuje, je v souladu s ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství v oblastech:

- Směrnice pro nízkonapěťové aplikace (2006/95/ES).
Použitá norma: EN 61010-1: 2001.
- Směrnice pro elektromagnetickou kompatibilitu (EMC) (2004/108/ES).
Použitá norma: EN 61326-1: 2006.

DK Overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet SI 010 CNV som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Lavspændingsdirektivet (2006/95/EF).
Anvendt standard: EN 61010-1: 2001.
- EMC-direktivet (2004/108/EF).
Anvendt standard: EN 61326-1: 2006.

DE Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt SI 010 CNV, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmt:

- Niederspannungsrichtlinie (2006/95/EG).
Norm, die verwendet wurde: EN 61010-1: 2001.
- EMV-Richtlinie (2004/108/EG).
Norm, die verwendet wurde: EN 61326-1: 2006.

GR Δήλωση Συμμόρφωσης

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα SI 010 CNV, στα οποία αναφέρεται η παρούσα δήλωση, συμμορφώνονται με τις εξής Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΕ:

- Οδηγία χαμηλής τάσης (2006/95/ΕΚ).
Πρότυπο που χρησιμοποιήθηκε: EN 61010-1: 2001.
- Οδηγία Ηλεκτρομαγνητικής Συμβατότητας (EMC) (2004/108/ΕΚ).
Πρότυπο που χρησιμοποιήθηκε: EN 61326-1: 2006.

ES Declaración de Conformidad

Nosotros, Grundfos, declaramos bajo nuestra propia responsabilidad que el producto SI 010 CNV, al cual se refiere esta declaración, está conforme con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

- Directiva de Baja Tensión (2006/95/CE).
Norma aplicada: EN 61010-1: 2001.
- Directiva EMC (2004/108/CE).
Norma aplicada: EN 61326-1: 2006.

FR Déclaration de Conformité

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit SI 010 CNV, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

- Directive Basse Tension (2006/95/CE).
Norme utilisée : EN 61010-1: 2001.
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).
Norme utilisée : EN 61326-1: 2006.

IT Dichiarazione di Conformità

Grundfos dichiara sotto la sua esclusiva responsabilità che il prodotto SI 010 CNV, al quale si riferisce questa dichiarazione, è conforme alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

- Direttiva Bassa Tensione (2006/95/CE).
Norma applicata: EN 61010-1: 2001.
- Direttiva EMC (2004/108/CE).
Norma applicata: EN 61326-1: 2006.

HU Megfelelősségi nyilatkozat

Mi, a Grundfos, egyedüli felelősséggel kijelentjük, hogy a SI 010 CNV termék, amelyre jelen nyilatkozik vonatkozik, megfelel az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak:

- Kisfeszültségű Direktíva (2006/95/EK).
Alkalmazott szabvány: EN 61010-1: 2001.
- EMC Direktíva (2004/108/EK).
Alkalmazott szabvány: EN 61326-1: 2006.

PL Deklaracja zgodności

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby SI 010 CNV, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

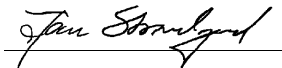
- Dyrektywa Niskonapięciowa (LVD) (2006/95/WE).
Zastosowana norma: EN 61010-1: 2001.
- Dyrektywa EMC (2004/108/WE).
Zastosowana norma: EN 61326-1: 2006.

RU Декларация о соответствии

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия SI 010 CNV, к которым относится настоящая декларация, соответствуют следующим Директивам Совета Евросоюза об унификации законодательных предписаний стран-членов ЕС:

- Низковольтное оборудование (2006/95/EC).
Применявшийся стандарт: EN 61010-1: 2001.
- Электромагнитная совместимость (2004/108/EC).
Применявшийся стандарт: EN 61326-1: 2006.

Bjerringbro, 24th June 2009



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Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

English (GB) Installation and operating instructions

Original installation and operating instructions.

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General description

Grundfos SI 010 CNV, sensor interface, is an external power supply, signal amplifier and signal converter for Grundfos sensors. SI 010 CNV has built-in precision resistors enabling the transmitter to give 1-5 V and 2-10 V output signals in stead of 4-20 mA.

2.1 Application

SI 010 CNV is used in sensor applications with controllers with 4-20 mA inputs in which the pressure and flow measurements are carried out by products from the Grundfos Direct Sensors™ standard product range.

2.2 Nameplate

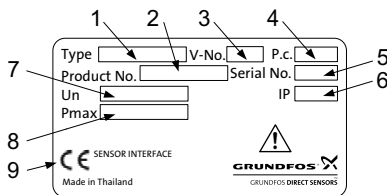


Fig. 1 Example of nameplate

Pos.	Description
1	Type designation
2	Product number
3	Version number
4	Production code (year and week)
5	Serial number
6	IP class
7	Supply voltage
8	Maximum input power
9	CE marking

2.3 Installation kit

The SI 010 CNV package contains these parts:

- SI 010 CNV unit
- installation and operating instructions for SI 010 CNV.

3. Configuration

Set the converter (SI 010 CNV) to the desired function before installing it. Carefully tip off the cover of the plastic cabinet using a small, flat screwdriver. See fig. 2.

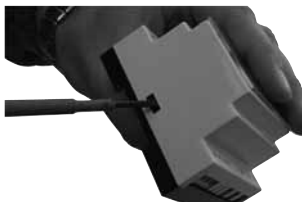


Fig. 2 Press in the tabs on both sides using a screwdriver

Inside the converter six small jumpers can be placed differently depending on the desired function. The short-circuit positions are designated ST1, ST2, ST3, ST4, ST5. See fig. 3.

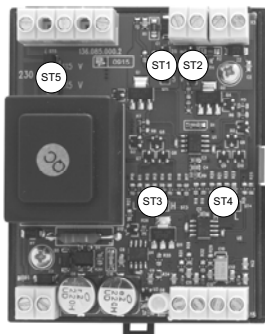


Fig. 3 Short-circuit positions on PC board

Figure 3 shows a picture of SI 010 CNV without cover. The circles, ST1-ST5, show the physical position of the short-circuit positions.



Warning

The SI 010 CNV must be configured before cables are connected as it is highly dangerous to configure the unit when power is connected.

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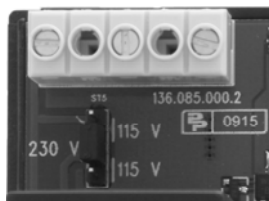
3.1 ST5 - supply voltage

Caution The SI 010 CNV can be connected to either 230/115 VAC or 24 VDC. It must NOT be connected to both voltages.

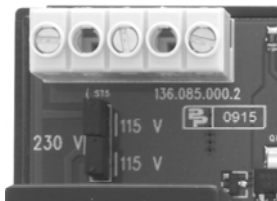
In position ST5, SI 010 CNV can be set to either 230 V or 115 V supply voltage:

- one jumper across the two pins in the middle = 230 V.
- two jumpers as shown in fig. 4 = 115 V.

If the converter is connected to 24 VDC, the configuration of ST5 has no function.



Set to 230 V (factory setting)



Set to 115 V

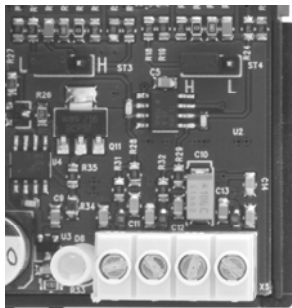
Fig. 4 How to set SI 010 CNV to 230 V and 115 V

3.2 ST3 and ST4 - sensor inputs

The sensors are available in two versions with different output signals, either 0.5 - 3.5 V or 0.5 - 4.5 V. The converter is compatible with both signal types which are configured in short-circuit positions ST3 and ST4.

Short-circuit position	Location (see fig. 7)	Function
ST3	L	Input 1, input voltage 0.5 - 3.5 V (factory setting)
	H	Input 1, input voltage 0.5 - 4.5 V
ST4	L	Input 2, input voltage 0.5 - 3.5 V (factory setting)
	H	Input 2, input voltage 0.5 - 4.5 V

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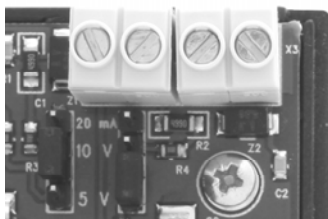


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Fig. 5 ST3 (input 1) is configured for sensors giving an 0.5 - 3.5 V signal, ST4 (input 2) is configured for sensors giving an 0.5 - 4.5 V signal

3.3 ST1 and ST2 - signal outputs

Short-circuit position	Location (see fig. 7)	Function
ST1	I	Output 1, output current 4-20 mA (factory setting)
	U1	Output 1, output voltage 2-10 V
	U2	Output 1, output voltage 1-5 V
ST2	I	Output 2, output current 4-20 mA (factory setting)
	U1	Output 2, output voltage 2-10 V
	U2	Output 2, output voltage 1-5 V



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Fig. 6 ST1 (output 1) is set to 2-10 V, and ST2 (output 2) to 1-5 V

3.4 Configuration examples

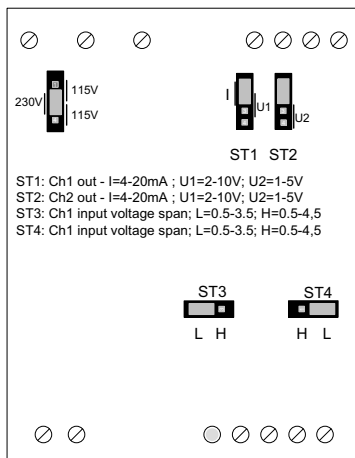
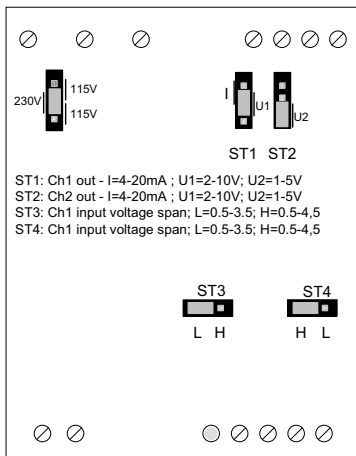


Fig. 7 Configuration from factory

Figure 7 shows the factory configuration (jumpers are shown as light grey rectangles): The supply voltage is set to 230 VAC (possibly 24 VDC), both sensor inputs are set to 0.5 - 3.5 V, and both signal outputs to 4-20 mA.

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Fig. 8 Alternative configuration

Figure 8 shows a configuration where the supply voltage is set to 230 VAC (possibly 24 VDC), sensor input 1 is set to 0.5 - 3.5 V, sensor input 2 is set to 0.5 - 4.5 V, signal output 1 is set to 2-10 V and signal output 2 to 1-5 V.

4. Installation

SI 010 CNV is designed for DIN rail mounting inside a control cabinet. An IP66 box for the mounting of SI 010 CNV is available as an accessory.

4.1 Mounting

When SI 010 CNV is mounted in a control cabinet also containing a frequency converter, it is especially important to ensure an EMC-correct installation of both SI 010 CNV and the frequency converter.

Consequently, the following rules must be observed:

- All cable screens must be connected to earth.
- All units in the control cabinet must be connected to earth.

Mounting on DIN rail

Use a 35 mm DIN rail (EN 50022).

1. Pull out the clip using a screwdriver (see fig. 9).
2. Mount SI 010 CNV on the DIN rail.
3. Make sure that the spring pushes back the clip so that SI 010 CNV is kept in place.

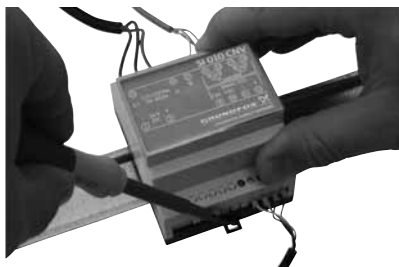


Fig. 9 Mounting SI 010 CNV on DIN rail

Dismounting from DIN rail

1. Pull out the clip using a screwdriver (see fig. 9).
2. Tip SI 010 CNV free of the DIN rail.

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4.2 Electrical connection

The electrical connection should be carried out by an authorised electrician in accordance with local regulations.



Warning

Before beginning the electrical connection, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

Caution

The supply voltage for SI 010 CNV can be 230/115 VAC 50/60 Hz, $\pm 10\%$ or 24 VDC (see section [4.2.1 Connection of supply voltage and sensor](#)).

Note

If the supply cable to SI 010 CN has no earth conductor, connect a separate earth connection.

Connect SI 010 CNV to 230/115 VAC, 50/60 Hz or 24 VDC. Never connect more than one voltage, and check with a voltmeter which voltage is available. Set the ST jumpers correctly. See section [3. Configuration](#).



Warning

Wrong connection may cause highly dangerous situations and destroy SI 010 CNV and possibly equipment connected to SI 010 CNV.

The voltage supply built into SI 010 CNV fulfils the PELV requirements, and the PE terminals (protective earth) and GND are connected.

Note

If SI 010 CNV is supplied from an external 24 V voltage supply, the supply **MUST** be PELV-approved.

Caution

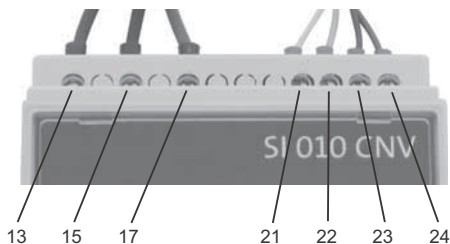
Connect SI 010 CNV to the public supply network via a switch and according to the local regulations. In order to ensure safety and correct function, it is important that the pipe system in which the sensor is mounted has an efficient earth connection.

Standard products from Grundfos Direct Sensors can be supplied via up to 2.9 m cables. See the installation and operating instructions of the sensor.

As far as possible, keep the connecting cables away from frequency converters and other noisy electronic equipment.

Use twisted-pair cable (minimum $2 \times 0.5 \text{ mm}^2$) between SI 010 CNV outputs and other equipment. Connect inputs and outputs according to the tables in section [4.2.1 Connection of supply voltage and sensor](#).

4.2.1 Connection of supply voltage and sensor



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Fig. 10 Electrical connection, earthing and connection of control box

Terminal	Description
13	Phase conductor 230/115 VAC (be sure to configure short-circuit position ST5 if it is a 115 V supply).
15	Neutral
17	Protective earth conductor (PE)
21	Output 1 (4-20 mA or 2-10 V depending on the position of the jumper in position ST1)
22	Earth
23	Output 2 (4-20 mA or 2-10 V depending on the position of the jumper in position ST2)
24	Earth

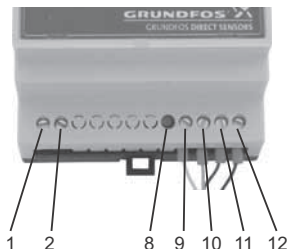


Fig. 11 24 VDC and sensor connection

Terminal	Description
1	External 24 VDC supply (PELV-approved)
2	Earth from external 24 VDC supply (PELV-approved)
8	Green diode which is on when SI 010 CNV is on
9	Sensor input 1 (yellow/white if a standard cable is used)
10	Sensor input 2 (yellow/white if a standard cable is used)
11	Earth for sensor (green if standard cable is used)
12	+ 5 V supply for sensor (brown if standard cable is used)

In some applications it may be required to measure both pressure and flow, but not temperature. The converter can supply two sensors and thus convert signals from them. However, this requires that two conductors are mounted in each of terminal 11 and 12. See fig. 12.

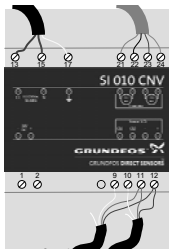


Fig. 12 Connection of two sensors

5. Technical data

Power supply

Voltage	230/115 VAC $\pm$ 10 % or 24 VDC
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Frequency	50/60 Hz
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Power consumption	Max. 2.5 W
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Cable dimension according to IEC	0.2 - 4 mm ² UL: 24-12 AWG
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Sensor connection

Recommended cable type	Screened Cross-section: 0.25 - 1 mm ² / AWG: 24-18 Max. cable length: 3 m (10 feet)
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Controller connection

Recommended cable type	Screened, twisted-pair 2-wire, 4-20 mA Cross-section: 0.26 - 1 mm ² / AWG: 23-17
------------------------	---

Max. cable length

Cable size			Max. cable length at 20 °C [m]	
AWG	Diameter [mm]	Cross-sectional area [mm ²]	Load resistance 500R	Load resistance 250R
17	1.15	1.04	8896	16309
18	1.02	0.82	7055	12934
19	0.91	0.65	5595	10257
20	0.81	0.52	4437	8134
21	0.72	0.41	3518	6451
22	0.64	0.33	2790	5116
23	0.57	0.26	2213	4057
24	0.51	0.20	1755	3217
25	0.45	0.16	1392	2551
26	0.40	0.13	1104	2023

Other data

Enclosure class	IP20
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Ambient temperature

During operation	-20 °C to +50 °C (-4 °F to +122 °F)
During storage	-20 °C to +70 °C (-4 °F to +158 °F)
During transport	-20 °C to +70 °C (-4 °F to +158 °F)

Caution

SI 010 CNV must not be exposed to direct sunlight during operation.

6. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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